Smart Prediction: Automated Period Matching & Model Selection

> ***Module***: `SpecialPeriod.py` *— Smart Predict Mode (Mode 2)*
> ***Purpose***: *Given an arbitrary date range, automatically identify its market regime, select the optimal model ensemble, and generate predictions.*

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1. Motivation

In the Classic Mode of `SpecialPeriod.py`, users must manually select models, markets, and periods. This requires domain expertise to know which models perform well under which market conditions. **Smart Prediction** automates this entire process:

- The user only provides a **start date**, **end date**, and **market**.
- The system determines what type of market regime the period resembles.
- It selects and weights the best-performing models accordingly.
- It trains, predicts, and outputs results — fully automated.

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2. System Architecture

```
User Input: start_date, end_date, market
                        |
      [Step 1] Feature Extraction
          Extract 6-dimensional statistical feature vector
          from actual market data within the date range
                        |
      [Step 2] Period Matching
          Compare against 7 predefined Special Periods
          using z-score normalized Euclidean distance
                        |
      [Step 3] Category Mapping
          Map matched period -> one of 4 regime categories
                        |
      [Step 4] Model Selection (per horizon)
          Use market-specific, horizon-aware rules
          to select models and ensemble weights
                        |
      [Step 5] Train & Predict
          Walk-forward training -> weighted ensemble prediction
                        |
      Output: predictions, curves, SmartPredict_Summary.csv
```

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3. Feature Extraction

For any given date range, the system extracts a **6-dimensional feature vector** from actual market price data:

#	Feature	Formula	Purpose
1	**Mean Return**	`mean(log_returns)`	Distinguish bull / bear / sideways
2	**Volatility**	`std(log_returns)`	Separate high-volatility crises from calm trends
3	**Max Drawdown**	`max(running_max(cum_ret) - cum_ret)`	Measure severity of decline
4	**Skewness**	`skew(log_returns)`	Left-skew = frequent sharp drops; right-skew = sharp rallies
5	**Kurtosis**	`kurtosis(log_returns)`	High kurtosis = fat tails / black swan events
6	**Trend Strength**	`sum(log_returns) / volatility`	Sharpe-like ratio; directional conviction

These features are computed from the **Log Return** column (first feature in the scaled data array), ensuring consistency with the training pipeline.

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4. Period Matching Algorithm

4.1 Reference Periods

The system maintains 7 predefined special periods as reference points:

Period	Date Range	Key Characteristics
2008 Financial Crisis	2007-10 ~ 2009-03	Systemic collapse, extreme volatility, deep drawdown
2010 Euro Debt Crisis	2010-04 ~ 2012-06	Prolonged moderate decline, repeated interventions
2015 China Crash	2015-06 ~ 2016-02	Sharp drop followed by rapid V-shaped rebound
2018 Trade War	2018-01 ~ 2018-12	Policy-driven, choppy, exogenous shocks
2020 COVID Crash	2020-02 ~ 2020-06	Black swan: fastest crash + fastest recovery
2022 Rate Hike	2022-01 ~ 2022-12	Fed tightening, persistent bear market
2023 AI Wave	2023-01 ~ 2023-12	Tech-driven sustained bull market

4.2 Distance Calculation

1. Compute the 6-dim feature vector for the user's input period.
2. Compute the 6-dim feature vector for each of the 7 reference periods.
3. Stack all 8 vectors (7 reference + 1 input) and apply **joint z-score normalization** (zero mean, unit variance per feature dimension).
4. Compute **Euclidean distance** between the normalized input vector and each reference vector.
5. Rank by distance — the closest reference period is the best match.

Joint normalization ensures that all features contribute equally regardless of their original scale.

4.3 Category Mapping

Each reference period maps to one of **4 regime categories**:

Category	Mapped Periods	Characteristics
`high_vol_crash`	2008 Financial Crisis, 2020 COVID	Extreme volatility, deep drawdown, systemic risk
`moderate_decline`	2010 Euro Debt, 2018 Trade War, 2022 Rate Hike	Gradual decline, choppy, policy-sensitive
`v_shape_recovery`	2015 China Crash	Sharp drop + rapid rebound, high drawdown but near-zero net trend
`trending_bull`	2023 AI Wave	Sustained uptrend, moderate volatility, strong positive trend

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5. Model Selection Strategy

5.1 Two-Level Strategy

The system employs a **dual strategy** with automatic fallback:

Strategy A — Historical Data (preferred):
If a `SpecialPeriod_Summary.csv` from prior experiments exists, the system:

1. Looks up the matched period's actual R^2 scores per model.
2. Filters by the current forecast horizon.
3. Excludes ARIMA (benchmark), SNN (unstable), and PatchTST (unless long-trending).
4. Selects the top-K models (default K=3) with $R^2 > 0$.
5. Assigns weights proportional to R^2 (with PatchTST penalized by 0.3x).

Strategy B — Heuristic Rules (fallback):
If no historical data is available, the system uses expert rules derived from comprehensive experiments on both SP500 and HSI markets.

5.2 Horizon-Aware Model Selection

Different forecast horizons require different model architectures. The system divides horizons into three bands:

Band	Horizon Range	Rationale
Short	$H \leq 1$	Next-day prediction; recent patterns dominate
Medium	$H = 2\sim5$	Short-term trend following; balanced complexity
Long	$H \geq 6$	Multi-day ahead; simpler models generalize better

Each horizon band has its own model selection and weight assignment, meaning a single Smart Predict run with H=1 and H=10 will use **different model ensembles** for each.

5.3 Market-Specific Rules

SP500 and HSI exhibit fundamentally different prediction characteristics. The system maintains **separate rule sets** for each market:

SP500 Rules

Based on SP500 analysis (Seed2620):

- **Overall DL ranking**: Mamba (0.64) > Bi-RNN (0.63) > Bi-GRU (0.58) > Bi-LSTM (0.48)
- Wide performance spread — weight differentiation matters
- PatchTST is catastrophic (R^2 = -0.80), always excluded
- SNN is unstable (R^2 = -0.13), always excluded

Category	Short (H≤1)	Medium (H=2~5)	Long (H≥6)
`high_vol_crash`	Mamba 0.30, Bi-GRU 0.25, Bi-RNN 0.25, Bi-LSTM 0.20	Bi-RNN 0.30, Mamba 0.25, Bi-LSTM 0.25, Bi-GRU 0.20	Mamba 0.40, Bi-RNN 0.30, Bi-GRU 0.30
`moderate_decline`	Mamba 0.30, Bi-GRU 0.25, Bi-RNN 0.25, Bi-LSTM 0.20	Mamba 0.30, Bi-RNN 0.25, Bi-LSTM 0.25, Bi-GRU 0.20	Mamba 0.40, Bi-RNN 0.30, Bi-GRU 0.30
`v_shape_recovery`	Mamba 0.35, Bi-GRU 0.25, Bi-RNN 0.20, Bi-LSTM 0.20	Bi-GRU 0.30, Bi-LSTM 0.25, Mamba 0.25, Bi-RNN 0.20	Mamba 0.40, Bi-RNN 0.30, Bi-GRU 0.30
`trending_bull`	Mamba 0.40, Bi-GRU 0.35, Bi-RNN 0.25	Bi-GRU 0.35, Mamba 0.35, Bi-RNN 0.30	Mamba 0.40, Bi-GRU 0.30, Bi-RNN 0.30

Key SP500 patterns:

- Mamba dominates Short horizon (led in 4/7 periods at H=1)
- Mamba leads Long horizon across all categories
- Bi-GRU is strong in trending/recovery markets

HSI Rules

Based on HSI analysis (Seed2620):

- **Overall DL ranking**: Bi-GRU (0.81) > Bi-RNN (0.81) > Mamba (0.80) > Bi-LSTM (0.80)
- Extremely tight top-4 cluster (only 0.01 R^2 spread) — weights are more uniform
- PatchTST is positive (R^2 = 0.52) but still well behind top-4
- SNN is positive (R^2 = 0.36) but too unstable

Category	Short (H≤1)	Medium (H=2~5)	Long (H≥6)
`high_vol_crash`	Bi-RNN 0.30, Bi-GRU 0.25, Bi-LSTM 0.25, Mamba 0.20	Bi-RNN 0.30, Mamba 0.25, Bi-LSTM 0.25, Bi-GRU 0.20	Bi-LSTM 0.40, Bi-RNN 0.30, Bi-GRU 0.30
`moderate_decline`	Bi-RNN 0.30, Bi-GRU 0.30, Mamba 0.20, Bi-LSTM 0.20	Bi-RNN 0.30, Bi-GRU 0.28, Mamba 0.22, Bi-LSTM 0.20	Bi-GRU 0.35, Bi-RNN 0.35, Mamba 0.30
`v_shape_recovery`	Bi-GRU 0.30, Mamba 0.25, Bi-LSTM 0.25, Bi-RNN 0.20	Bi-LSTM 0.30, Bi-GRU 0.25, Mamba 0.25, Bi-RNN 0.20	Bi-GRU 0.40, Bi-RNN 0.30, Bi-LSTM 0.30
`trending_bull`	Mamba 0.30, Bi-GRU 0.25, Bi-LSTM 0.25, Bi-RNN 0.20	Bi-GRU 0.28, Mamba 0.28, Bi-LSTM 0.22, Bi-RNN 0.22	Mamba 0.35, Bi-GRU 0.35, Bi-RNN 0.30

Key HSI patterns:

- Bi-RNN and Bi-GRU dominate Short horizon (led in 5/7 periods at H=1) — opposite of SP500
- Bi-LSTM is much stronger on HSI than SP500 (+0.32 R^2 improvement)
- Mamba only leads in `trending_bull` — unlike its broad SP500 dominance
- Weights are more uniform due to the tight performance cluster

5.4 PatchTST Inclusion Rule

PatchTST is **excluded by default** due to poor SP500 performance (R^2 = -0.80). However, it is **re-added** when:

1. The input period spans > **3 years**, AND
2. The matched category is a **trending** type (`trending_bull`, `high_vol_crash`, or `moderate_decline`)

When included, PatchTST receives a fixed weight of **0.20**, and all other weights are re-normalized. **Rationale**: PatchTST's patch-based attention mechanism can capture long-range patterns in sustained directional markets. On HSI it achieves positive R^2 (0.52), and even on SP500 it shows marginal positive R^2 during the 2023 AI Wave (a strong trending period). For short or volatile periods, its fine-grained temporal resolution loss is fatal.

5.5 Permanently Excluded Models

Model	Reason
ARIMA	Statistical benchmark only; should not participate in ensemble prediction
SNN	Extreme instability (SP500 R^2 range: -1.51 to 0.56; HSI std = 0.33), not reliable

6. Ensemble Method

6.1 Weighted Average Ensemble

The final prediction is a **weighted average** of individual model predictions:

$$\hat{y}_{ensemble} = \frac{\sum_{i=1}^{K} w_i \cdot \hat{y}_i}{\sum_{i=1}^{K} w_i}$$

Where:

- w_i = weight assigned to model i (from heuristic rules or historical R^2)
- $\hat{y}_i$ = prediction from model i
- K = number of selected models (typically 3–4)

6.2 Why Weighted > Equal-Weight

The original Classic Mode uses a naive equal-weight ensemble across all models, which **drags performance down** when weak models (PatchTST, SNN) are included. The SP500 experiment showed the Ensemble (R^2 = 0.57) underperforming both Mamba (0.64) and Bi-RNN (0.63) due to this effect.

Smart Prediction solves this by:

1. **Excluding** known poor performers entirely.
2. **Weighting** models proportionally to their expected performance.
3. **Adapting** weights per horizon and per market.

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7. Walk-Forward Training Protocol

Smart Prediction uses the same walk-forward methodology as Classic Mode:

```
|<------- Training Data ------->|<-- Period -->|
     All data before period start   User's input period


  Training set:  [data_start, period_start)
  Validation:    Last 10% of training (for early stopping)
  Test set:      Period data (with seq_len lookback)
```

- Models are trained **only on data before the period starts** — no future data leakage.
- Early stopping with patience=15 prevents overfitting.
- Each (window_size, horizon) combination trains fresh models independently.

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8. Usage

8.1 Running Smart Predict

```
python SpecialPeriod.py
```

Select **Mode 2** when prompted:

```
========== Mode Selection ==========
    1. Classic Mode (manual selection of models/periods)
    2. Smart Predict (auto-match period & select best models)
  ====================================
  Enter mode (1 or 2): 2
```

Then provide:

1. Market (SP500 or HSI)
2. Start date (YYYY-MM-DD)
3. End date (YYYY-MM-DD)
4. Sequence lengths (window sizes)
5. Forecast horizons

8.2 Output

The system outputs:

- **Feature vector**: 6-dim statistics of the input period
- **Similarity ranking**: Distance to all 7 reference periods
- **Best match**: Closest period and its category
- **Model selection**: Per-horizon model list with weights and strategy used
- **Prediction results**: MAE, R^2, RMSE for each individual model and the weighted ensemble
- **Prediction curves**: Actual vs predicted plots for each model and the ensemble
- **Summary CSV**: `SmartPredict_Summary.csv` with all results

8.3 Example Output

```
[FEATURES] Input period feature vector:
    MeanReturn     : -0.001234
    Volatility     : 0.023456
    MaxDrawdown    : 0.187654
    Skewness       : -0.543210
    Kurtosis       : +2.345678
    TrendStrength  : -1.234567

  [MATCH] Period similarity ranking:
    1. 2020_COVID_Crash              distance=1.2345 <<<
    2. 2008_Financial_Crisis         distance=2.3456
    3. 2015_China_Crash              distance=3.4567
    ...

  [RESULT] Best match: 2020_COVID_Crash (distance=1.2345)

    --- W=60 | H=5 ---
    [MODELS] Strategy: heuristic | Horizon band: medium
      Bi-RNN      : weight=0.300
      Mamba       : weight=0.250
      Bi-LSTM     : weight=0.250
      Bi-GRU      : weight=0.200
      Bi-RNN      : MAE=0.0123 | R2=0.8456
      Mamba       : MAE=0.0134 | R2=0.8234
      ...
      Ensemble    : MAE=0.0118 | R2=0.8567 (weighted)
```

9. Design Decisions & Rationale

Decision	Rationale
6-dim features instead of raw prices	Invariant to price level; captures regime characteristics (volatility, trend, tail risk)
Joint z-score normalization	Ensures all features contribute equally to distance; prevents volatility from dominating
Euclidean distance over cosine similarity	We care about magnitude differences (a period with 2x volatility IS different), not just directional similarity
4 categories not 7	Similar periods share optimal model configurations; grouping reduces overfitting to specific historical episodes
Per-horizon model selection	The SP500 analysis showed R^2 drops 83% from H=1 to H=10, with completely different best models at each horizon
Separate SP500/HSI rules	HSI's top-4 DL models are within 0.01 R^2 of each other, while SP500 has 0.16 spread — requiring fundamentally different weighting strategies
PatchTST >3yr threshold	PatchTST needs long-range patterns to function; short volatile periods destroy its patch-based attention
Weighted over equal ensemble	Prevents weak models from dragging down the ensemble; the Classic Mode's Ensemble underperformed individual models due to PatchTST/SNN contamination

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Report generated for SpecialPeriod.py Smart Prediction Module.